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SURGE PROTECTION

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

The purpose of this chapter is to highlight SEC practices with respect to the requirements and application of surge arrester.

2.0 GENERAL

Surge arresters are the primary protection to make the circuit devices immune to different types of overvoltages and spikes either atmospheric or switching. Surge arresters are designed to limit the voltage surges by discharging or bypassing surge current and automatically limiting the flow of system power current.

3.0 TYPES OF SURGE PROTECTIVE DEVICES

Two types of surge protective devices are generally provided:

3.1 Rod Gap

A rod gap is a spark gap comprising of two rods spaced coaxially, extending from their supports at least one half of the gap spacing. Spark over occurs when the air is ionized by a sufficient voltage potential across the rods. Disadvantages of rod gaps are, it conducts at higher voltages and requires very steep rise time. Further it has unpredictable turn-on and response characteristics.

Rod gaps shall be provided with prior SEC approval, only when found necessary for insulation co-ordination when the phase to ground insulation distance of the disconnect switch is 90% or more than the disconnect switch isolating distance. The design of rod gaps for such applications shall be the responsibility of the disconnect manufacturer.

3.2 Metal Oxide Surge Arrester

Metallic oxides surge arresters are ceramic semiconductor devices with highly non-linear current-voltage characteristics and with an extremely good energy capability. The voltage-current characteristics for the resistive components (I_r) and capacitive components (I_c) currents through metal oxide discs are shown in figure 06-1. The curve comprises of three regions namely pre-breakdown region (Region 1), breakdown region (Region 2) and high current region (Region 3). In the pre-breakdown region the resistivity of material depends on the temperature with negative temperature co-efficient, in the breakdown region the temperature dependence is very small and in the higher current region the characteristics are no longer linear. For surge arresters capacitive peak leakage current is normally 0.2mA to 3mA and resistive peak leakage current is 5% to 20% of capacitive leakage current under normal operating condition.

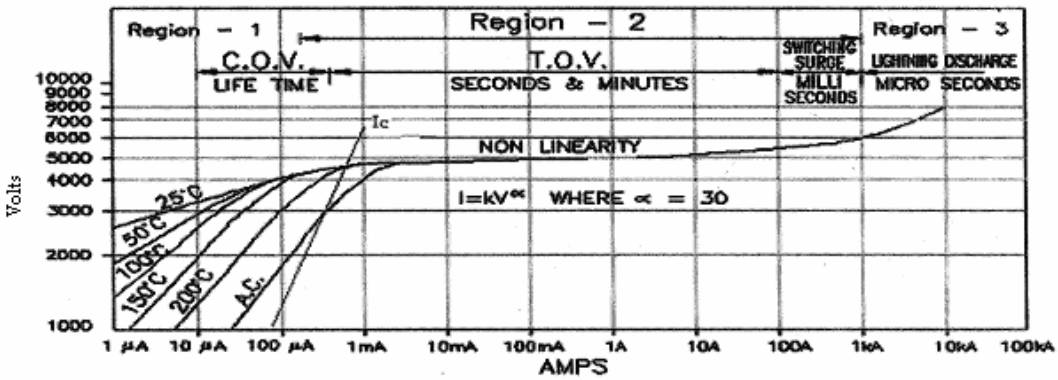

Figure 06-1 : Typical Voltage-Current Characteristics for Metal Oxide Arrester Disc

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4.0 ARRESTER SELECTION

The objective of arrester selection is to select the lowest rated surge arrester that will have a satisfactory service life on the power system while providing adequate protection of equipment insulation. An arrester of the minimum practical rating is generally preferred because it provides higher margin of protection for insulation. The use of higher rating increases the capability of the arrester to survive on the power system, but reduces the margin of protection it provides for a specific insulation level. Thus arrester selection must strike a balance between arrester survival and equipment protection.

The following shall be considered during the selection of surge arrester:

- Determination of maximum continuous operating voltage with respect to system highest operating voltage
- Determination of rated voltage with respect to temporary overvoltages
- Line discharges in switching surges
- Ambient temperature
  Co-ordination of protection level of the arrester with equipment to be protected
- Selection of pressure relief requirements for the arrester with respect to system short circuit currents
- Any special consideration

4.1 Maximum Continuous Operating Voltage (MCOV)

It is the maximum permissible r.m.s power frequency voltage applied continuously between the arrester terminals. MCOV rating for effectively grounded system shall be equal to or more than the system maximum line to ground power frequency voltage under emergency operating conditions and for non-effectively grounded system it shall be equal to or more than the system maximum line to line power frequency voltage under emergency operating conditions.

The recommended arrester MCOV capability is indicated in Table 06-1. MCOV rating of the arrester is normally about 80% of arrester rated voltage and it is the maximum permissible r.m.s value of the power frequency voltage between the arrester terminals at which it is designed to operate correctly as established in operating duty test.

4.2 Temporary Power Frequency Overvoltages

Temporary overvoltages (TOV) consist of lightly damped power frequency voltage oscillations, often with harmonics, usually lasting a period of hundreds of milliseconds or longer. Situations that may give rise to these overvoltages include single line-to-ground faults, ferro-resonance, load rejection, loss of ground, long unloaded transmission lines (Ferranti rise), coupled-line resonance and the transformer in-rush.

To determine the temporary overvoltage magnitude during fault conditions the system coefficient of grounding must be determined. The coefficient of grounding is defined the ratio of phase-to-ground power frequency voltage on a sound phase during fault at selected location to line-to-line power frequency voltage at the selected location with the fault removed. A system is considered effectively grounded if the coefficient of grounding does not exceed 80 percent, which is generally the case when system parameters are such that $0 < X_0/X_1 < 3$ and $0 < R_0/X_1 < 1$.

For non-effectively grounded system, during the ground fault the line-to-ground voltage in the other two healthy phases may rise up to system maximum line-to-line power frequency voltage under emergency operating condition for extended period. The maximum temporary voltage values for 10 seconds for effectively earthed system and non-effectively earthed system are shown in Table 06-1

Under normal operating condition there shall be a balance between the heat generated and dissipated by the arrester elements. However overvoltage events disturb these stable conditions by causing the elements to absorb increased levels of energy for the time the overvoltage exceeds the normal operating voltage. The arrester shall be capable of absorping this energy with or without prior energy absorption due to transmission line discharges per 35-TMSS-01.

Manufacturer shall furnish TOV (power frequency versus time) curve for the arrester with and without prior energy clearly indicating the overvoltage withstand value at 1 second and 10 seconds and the curve shall cover minimum time range of 0.1seconds to 20 minutes. The basic requirement is the power frequency overvoltage withstand value for the arrester shall be higher then the system TOV values for 10 seconds as indicated in Table 06-1.

4.3 Switching Surges

Switching surges are caused by normal line switching, high-speed autoreclosing, out of phase switching, switching of cable circuits/capacitor banks/shunt reactors, circuit breaker restriking, load rejection and current chopping etc.

Surge arresters dissipate switching surges by absorbing thermal energy. The amount of energy is related to the prospective switching magnitude, its wave shape, the system impedance, circuit topology, the arrester voltage-current characteristics, and the number of operations. The selected arrester shall have an energy absorption capability higher than the energy associated with expected switching surges. The ability of metal oxide arrester to absorb system switching surges can be quantified in terms of energy. The units generally used in quantifying this capability are kilojoules per kV (kJ/kV) at arrester rated voltage. Arrester energy absorption capability shall be per 35-TMSS-01.

4.4 Ambient Temperature and Solar Radiation

Higher temperature and aging of the material increases the energy of electrons in pre-breakdown region (refer figure 06-1), causing thermal breakdown. Hence performance and life expectancy of the arrester shall be based on the ambient conditions per 01-TMSS-01.

4.5 Co-ordination of Protective Level

Protective level of an arrester is the maximum crest voltage that appears across the arrester terminals under specified condition of operations. Protective level of an arrester comprising of lightning impulse protective level (LPL), and switching impulse protective level (SPL) shall be kept well below the equipment insulation level of basic lightning impulse insulation level (BIL) and basic switching impulse insulation level (BSL) respectively. Arrester protective level shall meet the requirements of Table 06-1.

4.6 Special Installation Conditions

If severe ambient temperature conditions exist for a particular installation, such as installation of a surge arrester inside an unventilated metal enclosure etc., these conditions shall be taken into consideration before selecting appropriate arrester.

5.0 ARRESTER APPLICATIONS

5.1 Power Transformers

5.1.1 Power Transformer HV and LV Winding

Arresters shall be located at or near all substation power transformers on both the high and low voltage sides except in cases where the transformer bushings are housed in cable termination boxes. The arresters shall be mounted directly on the transformers, wherever possible. For transformer with bushings terminating directly on to a Gas Insulated Switchgear (GIS), GIS surge arresters shall be provided at the termination point if required per insulation co-ordination studies.

The application of arresters for the protection of power transformers consists of the following steps :

Step 1: Select arrester rating (preliminary).

Step 2: Determine arrester protection levels (residual voltage) :
- Protection level for a full impulse wave
- Protection level for a switching impulse (if applicable).

Step 3: Determine transformer withstand capability :
- Basic insulation level
- Basic switching insulation level.

Step 4: Plot arrester protection levels and transformer withstand capability on a common coordinate system as shown in Figure 06-2.

Step 5: Determine protection margins between the arrester and transformer
•

Step 6: If protection margins are not higher than the recommended, select the next lower arrester rating that is compatible with the normal operating voltage of the system and go to step 2. Otherwise, go to Step 7.

Step 7: Determine whether arrester has adequate energy absorption capability for the application.

The above procedure is pertinent only for application of surge arresters at or near the transformer.

5.1.2 Power Transformer Tertiary Winding

For large transformer with low voltage tertiary winding surges can be transferred through the transformer from one winding to another. The voltages transferred through the transformers are mainly fast-front or slow-front voltages. The transferred surge has usually both the capacitively and inductively transferred components, which superimpose to the power frequency voltage.

The capacitively transferred components lay typically in megahertz range and are seen first in the transferred surge. The inductively transferred component comes after capacitive. The magnitude of the transferred voltage depends on the construction of transformer. Manufacturer shall furnish calculation with methods for the transferred surges for the surge arrester sizing. It shall be noted that not all the construction features of transformer influencing the transferred voltages can be addressed in the calculations and calculations can be used only as a rough estimation of surge magnitudes. The most practical method to quantitative estimate of surge is to measure them by tests.

5.2 Overhead Line Terminals

Arrester shall be installed at the overhead line entrances to substations with rated voltages 230kV and above. For other voltages calculations shall be carried out to decide the requirements.

5.3 Underground Line Terminals

5.3.1 Arresters need not be installed in 11kV, 13.8kV, 33kV, 34.5kV and 69kV substation switchgear when the line entrances are underground cables that do not terminate at an overhead system.

5.3.2 On 110 kV and higher voltages, arresters shall be installed, if required per insulation co-ordination studies, at both ends of underground cable which connects two substations.

5.3.3 For all voltage levels, when the line entrances are underground cables that terminate at an overhead system, arresters shall be installed at the underground cable to overhead line transition point. Grounding of transition point arresters to the substation ground through low impedance path is to be ensured.

5.4 Shunt Capacitors

Arresters shall be provided for all shunt capacitors. Further detailed calculation regarding, MCOV, rated voltage, TOV, energy absorption capability etc., shall be submitted. While installing capacitor banks for voltage improvement, existing silicon-carbide gapped arrester, if any, shall be replaced with metal oxide surge arrester.

5.5 Gas Insulated Substation (GIS)

GIS in general better protected than open-air substations because they have surge impedances much lower than the overhead lines. GIS with over head line connection through SF_6 to air bushing shall be provided with surge arrester at the air to GIS transition point. GIS with underground cables and over head line portion shall be provided with surge arresters per clause 5.3.3. For underground cables with rated voltage 110kV and above, connecting GIS of one substation to another substation GIS, surge arrester shall be provided if required per insulation co-ordination studies. For transformer connected by SF_6 bus duct additional surge arrester at the transformer or at the bus may be required when the separation distance to line entrance arresters is too long or when high voltages at the transformer are expected. Insulation co-ordination shall be carried out to decide the requirement of surge arresters and their location.

5.6 Shunt Reactors

Chopping of current and subsequent reignition by the shunt reactor breaker results in significant overvoltages. Surge arrester shall be provided for suppression of peak overvoltages.

Table 06-1 : Recommended Arrester Ratings

System Nominal Voltage	System Maximum Voltage under Emergency Operating Conditions for 30 minutes		Effectively Grounded System					Noneffectively Grounded System				
			Max. Temp. Over-voltage (10sec)	MCOV Rating	Arrester Rating	Maximum Residual Voltage		Max Temp Over-voltage (10 sec)	MCOV Rating	Arrester Rating	Maximum Residual Voltage	
						Lightning Impulse	Switching Impulse				Lightning Impulse	Switching Impulse
(kV_{LL})	(kV_{LL})	(kV_{LG})	(kV)	(kV)	(kV)	(kV_{peak})	(kV_{peak})	(kV)	(kV)	(kV)	(kV_{peak})	(kV_{peak})
1	2	3	4	5	6	7	8	9	10	11	12	13
11	12.1	7	10.5	8.4	10	32	25	13	10.2	12	35	32
13.8	15.2	8.8	12.5	10	12	38	30	16	12.7	15	42	39
33	36.3	21	32	24	30	90	70	38.5	29	36	100	94
34.5	38	22	32	24	30	90	70	41	31.5	39	105	90
69	75.9	43.8	62	48	60	175	130	76	57	72	210	170
110	121	70	100	76	96	250	200	128	98	120	320	250
115	127	73.3	114	84	108	300	230		.	.	.	.
132	145.2	83.8	125	92	120	330	260	142	106	132	350	280
230	253	146.1	203	152	192	500	400	.	.	.	.	.
380	418	241.3	375	265	360	950	800	.	.	.	.	.

6.0 OTHER CONSIDERATIONS

6.1 Effect of Lead Wires

The discharge of lightning currents through the inductance of connected lead wires produces a voltage that adds to the arrester discharge voltage. Lead length includes the ground lead length as well as the primary lead. The total length of these leads is measured from the point at which the arrester line connection is made to the point where interconnection is made between the arrester ground lead and the protected equipment ground lead, excluding the arrester length.

The minimum protective level margin between the equipment and voltage due to arrester discharge voltage plus voltage drop in the lead wire shall be 20%. Minimum cross-section of lead wire between the line connection and arrester terminal shall be per Table 06-2. As far as possible surge arrester shall be mounted nearest to the equipment to be protected so that the distance of the equipment is within the arrester's protective zone.

Table 06-2 : Minimum Sizes of Arrester Lead Wire

System Nominal Voltage (kV)	Lead Wire Minimum Size (mm^2)
11	70
13.8	70
33	70
34.5	70
69	120
110	120
115	120
132	120
230	240
380	240

Note : Sizes indicated in the above table are minimum and the actual size shall be based on voltage drop.

6.2 Mechanical Loading

The arrester shall not be used as a bus support device. The maximum permissible mechanical loading applied to a metallic-top arrester shall be 40 percent of manufacturer's rated cantilever strength.

The line connection on a porcelain-top arrester (ideally suited for confined spaces, such as cubicle mounting, where clearance between live parts are critical) shall be made in such a way that no excessive mechanical stress is placed on the arrester.

6.3 Grounding

The lead from the ground terminal of the arrester shall be connected by the shortest possible route to the insulated bushing of discharge counter and leakage current indicator (if provided) by a conductor with insulation rated not less than 5 kV crest and shall not be more than 3 meters in length, in any case, to avoid sparkover to ground and thereon directly to a dedicated ground rod via two number 240mm^2 stranded copper conductor, which shall also be connected to the station grid conductor in the substation. The protected transformer tank shall also be connected to the same grounding system as the arrester with as low ground resistance as possible, one ohm or less.

The arrester shall be insulated from ground with 5kV rated standoff insulators, so that the leakage current passes through the discharge counter and leakage current indicator.

6.4 Distance to Surrounding Objects

To prevent outside objects affecting the voltage grading, manufacturer shall furnish minimum distance for the objects (conducting and non-conducting type) to be located from the arrester.

If the distances stated are not maintained, there is a risk that the value of the discharge voltage of the arrester may differ. Also, the arrester may fail due to unbalanced stresses on it from nearby conducting or grounded objects.

6.5 Discharge Counters and Leakage Current Indicators

All arresters for system voltages of 69 kV and above shall have a discharge counter and a continuous AC leakage/internal current indicator. As contamination increases on the porcelain, the leakage current increases. The level of current registered serves as a guide to insulation cleaning frequency and abnormally high reading of counter over a given period of time would indicate the need for inspection and checking of arrester.

Due to the high contamination level and humid atmospheric conditions in SEC operating areas, an ammeter with a scale range of 0-50mA Peak / $\sqrt{2}$ (nonlinear scale) shall be provided to monitor and measure the leakage current across the external porcelain insulator, in addition to the internal current flowing through the arrester. The surge counter shall be non-resettable and capable of registering up to 5 discharges per second on a five digit cyclometer dial.

The discharge counter and the leakage current indicator is to be connected between the arrester ground terminal and the station ground and shall be installed in such a position that reading can easily be taken.

6.6 Pressure Relief Device

Arrester pressure relief device is intended to vent internal arc gases, and prevent violent porcelain shattering when an internal fault occurs. An arrester that has vented shall be replaced immediately. The arrester pressure relief class to be selected for a given application shall have a pressure relief capability greater than the maximum short circuit current available at the intended arrester location.

6.7 Shielding

Arrester discharge current is less than direct stroke current and the provision of an arrester is meant basically to protect substation equipments by limiting overvoltages in the form of traveling waves due to direct strokes entering the substation over connecting lines. Only in the case of a direct stroke very near to the terminal of arrester where there are no flashovers before arrester operation, the arrester is called upon to discharge most of the direct stroke lightning current. The possibility of such a situation shall be reduced by shielding the line and substation.

6.8 Grading Ring

Performance of zinc oxide arrester depends on protective levels, temporary overvoltage and energy discharge capabilities. Temporary overvoltage and energy handling capabilities are closely related to temperature of zinc oxide blocks, which in turn depends on power losses

Power losses increase rapidly with voltage stress and even distribution of voltage stress depends on influence of stray capacitance. For short arrester even distribution of voltage stress is possible as the self capacitance of zinc oxide block is higher than stray capacitance. However for arrester with longer height, influence of stray capacitance makes the voltage distribution less linear. The solution to this problem is by using external grading rings. Grading ring shall be provided for 380kV system arresters. For other voltages if required calculation shall be provided to prove even distribution of voltage.

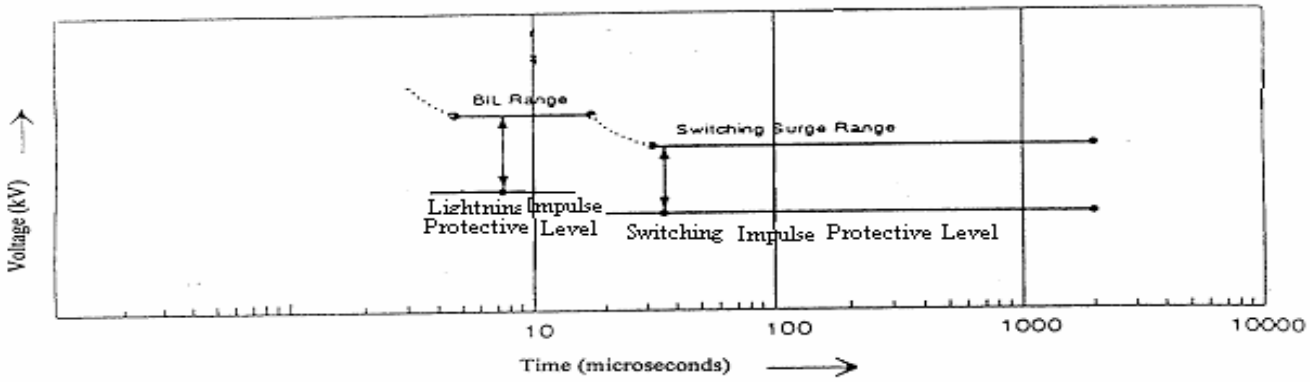

Fig 06-2 : Application of Surge Arresters for Protection of Power Transformers

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7.0 BIBLIOGRAPHY

1. ANSI C62.22, " Guide for the Application of Metal Oxide Surge Arresters for Alternating Current Systems "

2. IEC 60099-5, Surge Arresters-Part 5: "Selection and Application Recommendations".

3. IEEE Transactions on Power Apparatus and Systems, Vol. PAS-100, No.5, "Thermal Stability of the Gapless Surge Arrester " , 1981 .

4. IEEE Transactions on Power Apparatus and Systems, Vol. PAS-101, No.6, " Application of Metal Oxide Varistor Surge Arresters on Distribution Systems " , 1982 .

5. REA Bulletin 65-1, "Design Guide for Rural Substations", U.S. Department of Agriculture, 1978 .

6. Westinghouse Electric Corporation, "Electrical Transmission and Distribution Reference Book", Fourth Edition, Ninth Printing, Pennsylvania, USA, 1964 .

7. Westinghouse Electric Corporation , SA 252 , "Surge Protection of Power Systems ".